

SANDBERG, CA, USA

WMO: 723830

Lat: 34.744N

Lon: 118.724W

Elev: 1375

StdP: 85.86

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23187

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.0	-2.1	-21.2	0.7	8.3	-18.4	0.9	8.0	17.9	2.5	15.9	3.2	6.9	340	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	33.9	16.1	32.4	15.6	31.1	15.1	18.7	29.2	17.6	28.5	16.6	27.8	5.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	13.2	22.2	13.4	11.4	22.2	12.1	10.4	21.7	58.5	29.1	54.5	28.7	51.3	27.7	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.9	12.8	11.5	DB	-5.4	36.4	2.0	1.7	-6.8	37.6	-7.9	38.6	-9.1	39.5	-10.5	40.8	(4)
(5)			WB	-7.2	20.7	2.1	3.2	-8.6	23.0	-9.9	24.9	-11.0	26.8	-12.6	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	7.2	6.8	9.1	10.7	14.8	20.1	24.2	24.3	21.5	15.9	10.6	6.6	(6)	
	DBStd	8.05	4.41	4.87	5.09	5.46	5.34	5.11	3.31	3.32	4.68	5.19	5.18	4.34	(7)	
	HDD10.0	571	109	115	80	58	17	2	0	0	1	12	54	122	(8)	
	HDD18.3	2109	347	324	287	235	137	41	2	3	26	109	236	363	(9)	
	CDD10.0	2163	21	25	53	78	164	306	442	444	344	193	73	18	(10)	
	CDD18.3	659	0	1	2	5	27	94	186	189	120	32	4	0	(11)	
	CDH23.3	5481	0	2	7	28	192	828	1687	1679	888	162	7	0	(12)	
	CDH26.7	2079	0	0	0	2	48	307	691	692	309	30	0	0	(13)	
(14) Wind	WSAvg	6.0	6.3	6.4	6.3	6.5	6.4	6.2	5.3	5.2	5.1	5.8	6.0	6.5	(14)	
(15) Precipitation	PrecAvg	368	71	102	60	23	9	2	3	2	5	17	30	55	(15)	
	PrecMax	756	309	367	210	76	97	28	31	5	25	111	82	204	(16)	
	PrecMin	142	0	0	2	1	0	0	0	0	1	0	7	0	(17)	
	PrecStd	196	86	95	50	22	19	6	7	1	6	24	20	54	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.7	22.2	24.3	26.4	30.5	35.5	35.9	35.6	33.7	29.6	24.1	19.5	(19)	
		MCWB	7.6	9.0	10.4	11.4	14.8	16.2	16.9	17.2	15.7	12.6	10.4	7.4	(20)	
	2%	DB	16.5	18.3	21.1	23.6	27.9	32.5	33.9	33.5	31.8	27.1	22.1	16.4	(21)	
		MCWB	6.5	7.7	9.5	10.6	13.7	15.2	16.3	16.0	14.9	12.3	9.1	6.4	(22)	
	5%	DB	14.6	16.0	18.7	21.2	25.4	30.1	32.2	32.1	30.2	25.1	19.6	14.3	(23)	
		MCWB	5.4	6.4	8.7	10.3	12.8	14.6	16.0	15.5	14.2	11.3	8.2	5.5	(24)	
	10%	DB	13.0	13.9	16.6	18.8	23.0	27.9	30.6	30.7	28.3	23.0	17.6	12.6	(25)	
		MCWB	4.8	5.4	7.6	9.3	12.0	14.3	15.5	15.0	13.7	10.6	7.4	4.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	10.4	11.8	14.8	18.0	20.9	20.7	20.2	18.9	16.0	12.2	11.0	(27)	
		MCDB	13.3	17.8	20.2	20.0	22.0	27.1	30.2	30.8	27.1	23.8	20.4	13.1	(28)	
	2%	WB	8.5	8.8	10.6	12.3	15.0	17.1	18.6	18.0	17.2	14.2	10.3	9.0	(29)	
		MCDB	12.5	15.5	18.8	19.8	24.9	29.4	29.9	29.6	26.2	22.5	18.6	12.4	(30)	
	5%	WB	7.4	7.6	9.6	11.0	13.8	15.9	17.7	16.9	16.1	12.8	9.4	7.7	(31)	
		MCDB	11.2	13.5	16.7	19.0	23.7	27.6	28.8	29.3	26.0	21.8	17.0	11.3	(32)	
	10%	WB	6.4	6.6	8.6	10.0	12.7	14.9	16.7	15.9	15.0	11.7	8.4	6.5	(33)	
		MCDB	10.5	11.8	14.9	17.6	21.9	26.2	27.6	28.4	25.6	20.7	15.3	10.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	7.1	8.4	9.4	10.3	11.2	11.3	11.4	10.7	9.1	7.6	6.8	(35)	
		MCDBR	7.9	9.3	10.2	11.4	12.0	12.4	12.0	11.7	11.3	10.3	8.8	8.7	(36)	
	5% WB	MCWBR	4.6	5.0	5.4	5.8	5.6	5.1	5.0	4.9	4.6	4.6	4.3	4.8	(37)	
		MCWBR	7.0	8.3	9.4	10.6	11.6	11.4	10.9	11.0	10.9	9.9	8.4	7.5	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.260	0.269	0.280	0.309	0.318	0.317	0.336	0.318	0.311	0.295	0.267	0.264	(40)	
		taud	2.611	2.608	2.569	2.474	2.472	2.480	2.461	2.489	2.511	2.547	2.611	2.609	(41)	
	Edn at Noon	Edn at Noon	969	993	1001	977	964	960	939	955	951	945	952	943	(42)	
		Edh at Noon	74	83	94	109	111	110	111	106	98	87	73	70	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	3.01	3.90	5.34	6.62	7.50	8.19	7.91	7.42	6.26	4.68	3.48	2.75	(44)	
	RadStd	RadStd	0.24	0.45	0.34	0.35	0.41	0.37	0.30	0.18	0.14	0.28	0.16	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.25	N/A	-1.69	+1.45	N/A	N/A	-120	-264	+342	+183	(46)
(47)	Regional (49 neighbors)	+0.62	N/A	-1.65	+0.64	N/A	N/A	N/A	-121	+223	+93	(47)

Nomenclature: See separate page